

Ambient Assisted Living System

Daily Activity Report

Generated: Monday, 06 April 2026 at 22:15 | Total Sessions: 3 | UCI HAR Dataset

Session 1 | Dominant: Walking | Active 47.3% / Sedentary 52.7% [WARNING: ANOMALIES FLAGGED]

Session Metrics Summary

Metric	Value	Metric	Value
Total Sensor Readings	1000	Activity Transitions	97
Active Time	47.3% (473 readings)	Sedentary Time	52.7% (527 readings)
Dominant Activity	Walking	Unique Activities	6
Avg Model Confidence	96.1%	Low-Confidence Readings	26
Total Anomalies	61	Anomaly Rate	6.1%
Max Reconstruction Error	0.09361	Avg Reconstruction Error	0.03029

Activity-by-Activity Breakdown

Activity	Readings	Share	Avg Conf.	Anomalies	Category
Walking	202	20.2%	97.2%	8	Active
Laying	183	18.3%	100.0%	16	Sedentary
Sitting	179	17.9%	92.8%	7	Sedentary
Standing	165	16.5%	93.5%	5	Sedentary
Walking Upstairs	137	13.7%	93.5%	3	Active
Walking Downstairs	134	13.4%	99.5%	22	Active

Detailed Activity Narrative

Physical Health Observations

The resident's activity pattern suggests a reasonable level of mobility and endurance. The high number of walking activities indicates that the resident is physically active, which is beneficial for maintaining independence and overall health. The balance between active and sedentary time is relatively even,

which is appropriate for an elderly resident.

However, extended periods of daytime lying down may warrant caregiver attention, as it could potentially indicate fatigue or discomfort. It is crucial to ensure that the resident is getting up and moving around regularly to prevent potential health issues.

Anomaly and Sensor Confidence Analysis

There were 61 readings flagged as anomalous, which represents 6.1% of the total sensor readings. These anomalies could potentially reflect unusual movement, a potential health event, or sensor noise. The model's confidence level was 96.1%, with 26 low-confidence readings. Low-confidence readings might suggest that the sensor data is not as reliable, and it would be prudent to verify these readings with the caregiver's observations.

Anomalies were concentrated in the laying and walking down stairs activities, which could indicate unusual movements or potential health events during these activities.

Caregiver Recommendations for Tomorrow

To ensure the resident's health and safety, it would be beneficial to encourage more active periods throughout the day, particularly walking activities. This can help improve the resident's mobility, balance, and overall health.

It is also important to monitor the resident for any signs of fatigue or discomfort, especially during extended periods of rest. If the resident seems unusually tired or uncomfortable, it may be necessary to adjust the daily routine to allow for more rest periods.

Lastly, given the number of anomalous readings, it would be advisable to conduct a wellness check to ensure the resident's health and safety. This could involve a simple conversation about the resident's well-being, checking for any signs of discomfort or pain, and verifying the accuracy of the sensor data.

In terms of lifestyle adjustments, it may be beneficial to encourage the resident to break up extended periods of rest with short walks or stretching exercises. This can help improve the resident's mobility, reduce the risk of health issues associated with prolonged rest, and contribute to a more active and engaging daily routine.

Session 2 | Dominant: Standing | Active 48.0% / Sedentary 52.0%

Session Metrics Summary

Metric	Value	Metric	Value
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Total Sensor Readings	1000	Activity Transitions	71
Active Time	48.0% (480 readings)	Sedentary Time	52.0% (520 readings)
Dominant Activity	Standing	Unique Activities	6
Avg Model Confidence	96.8%	Low-Confidence Readings	13
Total Anomalies	35	Anomaly Rate	3.5%
Max Reconstruction Error	0.15285	Avg Reconstruction Error	0.02912

Activity-by-Activity Breakdown

Activity	Readings	Share	Avg Conf.	Anomalies	Category
Standing	199	19.9%	96.6%	4	Sedentary
Walking	193	19.3%	95.5%	7	Active
Laying	181	18.1%	99.4%	8	Sedentary
Walking Upstairs	147	14.7%	96.3%	1	Active
Walking Downstairs	140	14.0%	97.9%	14	Active
Sitting	140	14.0%	94.8%	1	Sedentary

Detailed Activity Narrative

Physical Health Observations

The activity pattern indicates that the resident is generally mobile and has good balance, as evidenced by the frequent walking and stair climbing activities. The resident's endurance is also reasonably good, as shown by the prolonged standing periods without movement. However, prolonged standing without movement can cause fatigue, so it is essential to encourage the resident to take breaks during such activities.

The balance between active and sedentary time is appropriate for an elderly resident, as it promotes overall health and well-being. The resident's level of movement is within the recommended daily activity levels for seniors, which emphasize a balance between physical activity and rest.

Anomaly and Sensor Confidence Analysis

There were 35 anomalous readings, which accounted for 3.5% of the total readings. These anomalies could potentially reflect unusual movement, a potential health event, or sensor noise. The model's confidence level was high, with an average of 96.8%, indicating that the readings were generally accurate. Low-confidence readings, below 60%, were concentrated in the walking and sitting activities,

which could suggest that the sensor may have difficulty detecting these activities accurately.

Caregiver Recommendations for Tomorrow

To encourage a healthy and active lifestyle for the resident, it is recommended that the caregiver encourage the resident to engage in a variety of activities, including walking, stair climbing, and seated exercises. It is also essential to remind the resident to take breaks during prolonged standing activities to prevent fatigue.

The caregiver should review the activity logs for any irregular movement bursts, as these could potentially indicate a health event. If the resident experiences any unusual movements or symptoms, a wellness check may be necessary.

To further improve the resident's health and safety, it is recommended that the caregiver encourage the resident to engage in activities that promote balance and strength, such as tai chi or yoga. Additionally, it is essential to ensure that the resident gets enough rest and sleep, as extended daytime laying may warrant caregiver attention.

Session 3 | Dominant: Standing | Active 46.3% / Sedentary 53.7% [WARNING: ANOMALIES FLAGGED]

Session Metrics Summary

Metric	Value	Metric	Value
Total Sensor Readings	947	Activity Transitions	58
Active Time	46.3% (438 readings)	Sedentary Time	53.7% (509 readings)
Dominant Activity	Standing	Unique Activities	6
Avg Model Confidence	97.6%	Low-Confidence Readings	5
Total Anomalies	52	Anomaly Rate	5.5%
Max Reconstruction Error	0.10472	Avg Reconstruction Error	0.02657

Activity-by-Activity Breakdown

Activity	Readings	Share	Avg Conf.	Anomalies	Category
Standing	176	18.6%	97.4%	1	Sedentary
Laying	173	18.3%	100.0%	3	Sedentary

Walking Upstairs	164	17.3%	97.7%	7	Active
Sitting	160	16.9%	96.4%	3	Sedentary
Walking Downstairs	139	14.7%	95.6%	22	Active
Walking	135	14.3%	98.3%	16	Active

Detailed Activity Narrative

Physical Health Observations

The resident's activity pattern suggests that they are generally mobile and have good balance, as evidenced by their engagement in various physical activities. However, prolonged standing without movement can cause fatigue, so it's essential to encourage the resident to take breaks during such activities.

Extended daytime laying may warrant caregiver attention, as it could indicate fatigue or discomfort. On the other hand, it's essential for recovery, especially after physical exertion. The resident's level of movement is appropriate for an elderly resident and aligns with recommended daily activity levels for seniors, which emphasize a balance between activity and rest.

Anomaly and Sensor Confidence Analysis

There were 52 readings flagged as moderate anomalies, which could potentially reflect unusual movement, sensor noise, or a potential health event. These anomalies were scattered across various activities, with walking on flat surfaces and walking downstairs having the most anomalous readings. It's crucial for the caregiver to review the activity logs for these irregular movement bursts to determine if they are cause for concern.

The model's confidence level was high, with an average of 97.6%. Low-confidence readings, which were below 60%, were rare, occurring only 5 times. These low-confidence readings might suggest sensor noise or a temporary glitch in the system.

Caregiver Recommendations for Tomorrow

To ensure the resident's continued health and safety, the caregiver should encourage activities that promote mobility, balance, and overall physical health. This could include short walks, seated exercises, or activities that encourage stair climbing.

The caregiver should also watch for warning signs such as prolonged periods of standing without movement, extended daytime laying, or irregular movement bursts. If these occur frequently or for extended periods, a wellness check may be necessary.

Lastly, lifestyle adjustments such as regular exercise, a balanced diet, and adequate rest can significantly improve the resident's health and safety. Encouraging the resident to maintain a routine that includes a balance of activity and rest is crucial for their overall well-being.
